

■ 3.4.10 Setting Up Algebraic Transformation Rules

Transformation rules in *Mathematica* are usually based on the *structure* of expressions, not their algebraic meaning. Using `Solve` and related functions, however, you can effectively carry out true algebraic transformations on expressions.

The transformation rule replaces the specific object `x^2` by `a`. The rule says nothing about objects like `x^4` that are algebraically related to `x^2`.

```
In[1]:= 1 + x^2 + x^4 /. x^2 -> a
```

```
Out[1]= 1 + a + x^4
```

By calling `Solve` with the equation `x^2 == a`, and asking it to eliminate `x`, you can effectively carry out a true algebraic transformation.

```
In[2]:= Solve[{ f == 1 + x^2 + x^4, x^2 == a }, f, x]
```

```
Out[2]= {{f -> 1 + a + a^2}}
```

Often, you need to perform the same algebraic transformation several times. *Mathematica* therefore allows you to construct a set of *algebraic rules*, which embody the transformations you specify, and which you can apply simply by using the standard `/. operator`.

<code>AlgebraicRules[eqns, vars]</code>	generate a set of algebraic rules which replace variables earlier in the list <code>vars</code> by ones later in the list, according to the equations <code>eqns</code>
<code>expr /. algrules</code>	apply algebraic rules to a particular expression

Algebraic transformation rules.

This sets up algebraic transformation rules which try to replace `x` by `a` according to the equation `x^2 == a`.

```
In[3]:= ar = AlgebraicRules[x^2 == a, {x, a}]
```

```
Out[3]= {x^2 -> a}
```

You can apply the algebraic rules using the `/. operator`.

```
In[4]:= 1 + x^2 + x^4 /. ar
```

```
Out[4]= 1 + a + a^2
```

The algebraic rules replace as many powers of `x^2` as possible.

```
In[5]:= 1 + x^3 + x^7 /. ar
```

```
Out[5]= 1 + a x + a^3 x
```

Algebraic rules in *Mathematica* are set up using mathematical objects known as *Gröbner bases*. Although the output from `AlgebraicRules` is usually displayed as a simple list of rules, it is in fact a representation of a Gröbner basis.

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This is the internal form of the object produced by `AlgebraicRules`. It is a representation of a Gröbner basis.

```
In[6]:= InputForm[ar]
Out[6]//InputForm=
AlgebraicRulesData[{x, a}, {x, a}, True,
  {{{2, {{0, 1}}}, {0, {{1, -1}}}}, {x, a},
  {x^2 -> a}, Rational]
```

Many of *Mathematica*'s internal equation manipulation algorithms use Gröbner bases in the form produced by `AlgebraicRules`. If you need the explicit form of a Gröbner basis, you can find it by pulling out the appropriate part of the expression you get from `AlgebraicRules`.

You can use `AlgebraicRules` to set up algebraic rules involving several variables.

This sets up algebraic rules which try to replace `x` and `y` by `a` and `b` using the equations given.

```
In[7]:= ar = AlgebraicRules[{a == x+y, b == x y},
  {x, y, a, b}]
Out[7]= {-y^2 -> b - a y, -x -> -a + y}
```

By applying the algebraic rules `ar`, you can rewrite `x^3 + y^3` in terms of `a` and `b`.

```
In[8]:= x^3 + y^3 /. ar
Out[8]= a^3 - 3 a b
```

You can apply the same rules to `x^4 + y^4`.

```
In[9]:= x^4 + y^4 /. ar
Out[9]= a^4 - 4 a^2 b + 2 b^2
```

In this case, it is not possible to eliminate `x` and `y` entirely. Following the order of variables you specified in `AlgebraicRules`, *Mathematica* nevertheless tries to eliminate `x` in favor of `y`.

```
In[10]:= x^3 + y^4 /. ar
Out[10]= a^3 - 2 a b - a^2 b + b^2 - a^2 y + a^3 y - 2 a b y +
  b y
```